

## A short derivation

A backslash breaks the block onto a new line; an ampersand marks the column everything should line up on. Here every line aligns on its equals sign:

$$\begin{aligned}(a + b)^2 &= (a + b)(a + b) \\ &= a^2 + ab + ba + b^2 \\ &= a^2 + 2ab + b^2\end{aligned}$$

The same trick keeps the steps of a solved equation stacked neatly:

$$\begin{aligned}3x + 6 &= 21 \\ 3x &= 15 \\ x &= 5\end{aligned}$$